

21.05.2026

Applied Machine Learning

HYPERPARAMETER OPTIMIZATION

- Quiz
- Assignment
- Projects
- Input
 - Performance Estimation
 - *k*-Fold Cross Validation
 - Early Stopping

QUIZ



<https://forms.office.com/e/hTSkh5QbHh>

ASSIGNMENT PRESENTATION

ASSIGNMENT: WHO PRESENTS NEXT?

PROJECTS

PROJECT MILESTONES

- 23.04. Present your Ideas
- 30.04. Form Groups & Put your Name in the [Excel](#)
- 07.05. Literature Review
- 14.05. Dataset Characteristics and Individual Feedback Sessions (**PUBLIC HOLIDAY**)

Deadline for completing the repo sections: 17.05.

- 21.05. Definition of Model Evaluation
- 28.05. Project Work
- 04.06. Baseline Model Estimation

Deadline for completing the repo sections: 07.06.

- 11.06. Project Work / Optional Feedback Sessions (**NO REGULAR CLASS**)
- 25.06. Project Presentations, Part I
- 02.07. Project Presentations, Part II

Submission deadline for the documented repo: 01.08.2026

PROJECTS: EVALUATION METRICS

- MSE
- MAE
- Accuracy
- F1 Score
- Recall
- Precision
- Confusion Matrix
- ...

PERFORMANCE ESTIMATION

PERFORMANCE ESTIMATION

Why would we need an additional test set, when we already have a validation set?

- Hyperparameter tuning is often done based on the validation set
- Validation set thus influences model choice/performance
- **Performance estimate is no longer unbiased** and overfitting may occur

What's the downside of splitting the data only once into training, validation, and test set?

- We use only one specific subset of the data to test on
- We could get a **better performance estimate** if we tested on more data

K-FOLD CROSS VALIDATION

K-FOLD CROSS VALIDATION

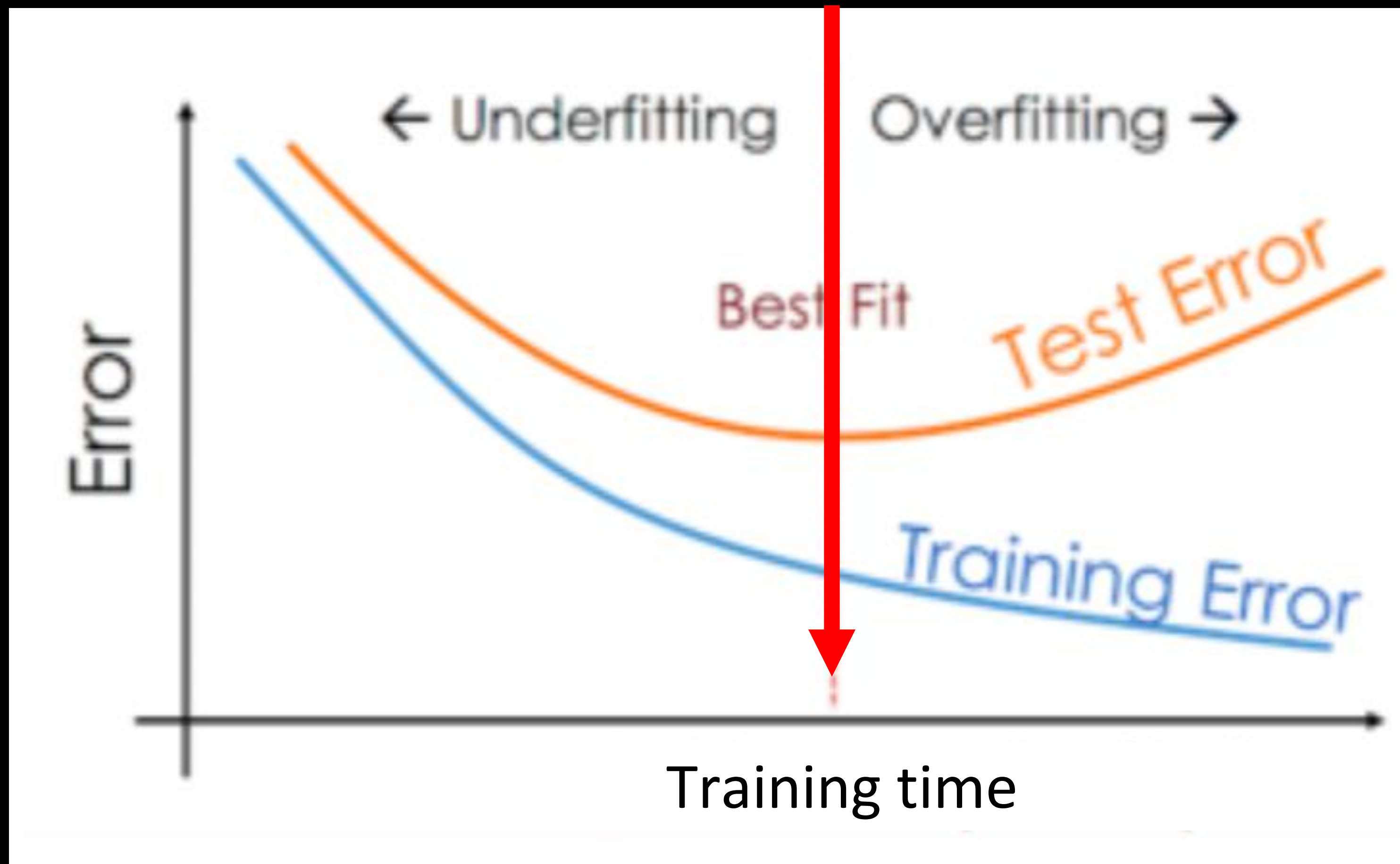
4-fold validation (k=4)



EARLY STOPPING

EARLY STOPPING

Early stopping



EARLY STOPPING

- Monitor **validation loss** during training
- If validation loss **does not decrease** (enough), stop training early
- **Patience**: validation loss might continue to decrease after a few epochs, so it may be worth continuing training for a few epochs
- **Restoring weights**: store the weights of the best model in a variable and restore them after training is finished